

Classification of Pedestrians-Cyclists Interactions in Shared Spaces

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About this document

What is it ?

A spatio-temporal classification of pedestrians-cyclists interactions (individual-individual, group-individual, group-group) in shared spaces.

How was it created ?

Built from the CTV dataset used as a benchmark for essential, diverse and controlled pedestrians-cyclists interactions (5009 interactions in total).

Additional interactions observed in real-world conditions (but absent from CTV or in a few quantity) were added to the classification.

N.B: This classification is not exhaustive as some interactions were not observed in CTV videos nor in real life conditions.

How to use it ?

If you have access to speed, direction and position information about a pedestrian-cyclist interaction, you can classify it with the table.

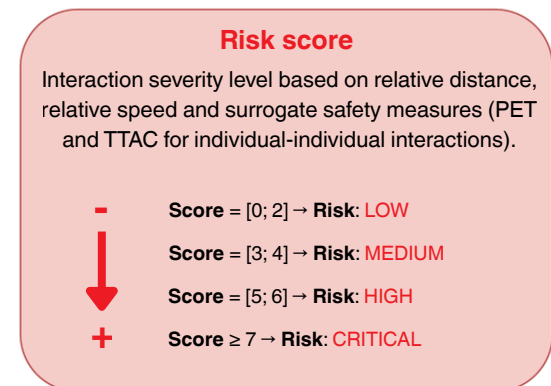
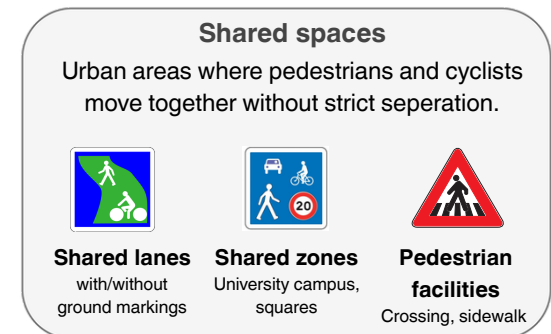
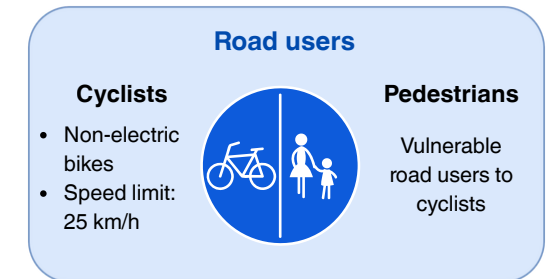
Why was it created ?

To classify as objectively as possible a wide variety of interactions between pedestrians and cyclists in shared spaces, and then to recognize them.

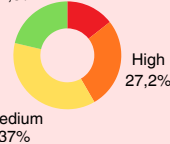
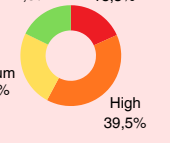
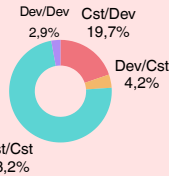
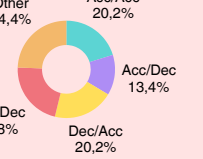
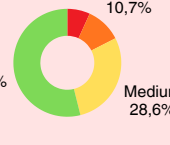
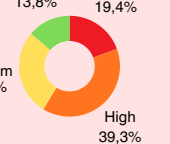
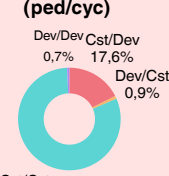
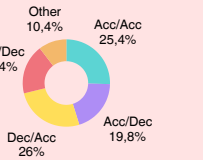
Context of creation ?

Internship about the analysis and simulation of pedestrian-cyclist interactions in shared spaces, at Cerema and IRIT, under the supervision of Josué Rivera and Manon Prédhumeau.

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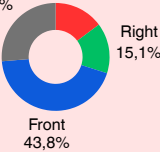
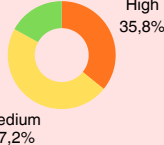
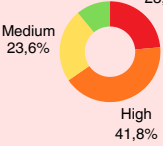
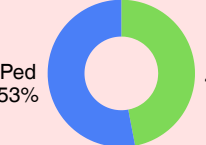
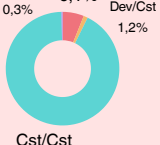
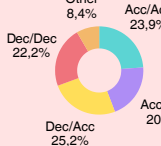
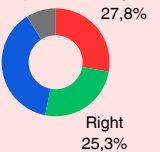
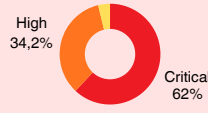
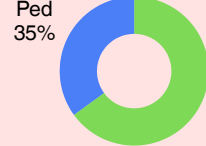
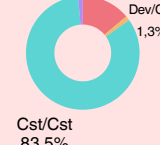
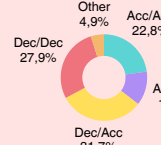


Individual-Individual Interactions

Icon	Class	Direction	Approach angle	Position	Average minimal distance	Average relative speed	Risk metrics	Reactivity statistics
	Overtaking (815 detected in CTV)	Same	Approach & Moving away	Sequence: Behind → Lateral → Front	2.52 m ± 1.18 m	9.31 ± 9.04 km/h 2.59 ± 2.51 m/s	PET-based risk  TTAC-based risk 	Spatial deviation responsiveness  Spatial deviation combinations (ped/cyc)  Speed variation combinations (ped/cyc) 
	Avoidance (864 detected in CTV)	Opposite	Approach & Moving away	Sequence: Front → Lateral → Behind	2.56 m ± 1.13 m	15.32 ± 5.67 km/h 4.26 ± 1.58 m/s	PET-based risk  TTAC-based risk 	Spatial deviation responsiveness  Spatial deviation combinations (ped/cyc)  Speed variation combinations (ped/cyc) 

Abbreviations:

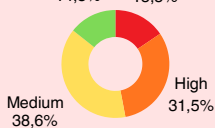
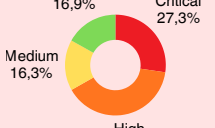
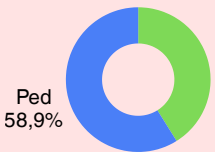
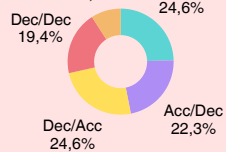
- Ped: Pedestrian
- Cyc: Cyclist
- Acc: Acceleration
- Dec: Deceleration
- Cst: Constant

Icon	Class	Direction	Approach angle	Position	Average minimal distance	Average relative speed	Risk metrics	Reactivity statistics
	Crossing (365 detected in CTV)	Crossing (intersection of paths)	Approach & Moving away (obliquely or not)	Sequence: Behind / Front → Lateral → Front Relative position 	3.11 m ± 1.02 m	14.18 ± 19.83 km/h 3.94 ± 5.51 m/s (standard deviation may be wrong due to data noise)	PET-based risk  TTAC-based risk 	Spatial deviation responsiveness  Spatial deviation combinations (ped/cyc)  Speed variation combinations (ped/cyc) 
	(Near) Collision (79 detected in CTV)	All	Approach (& moving away if avoidance)	Sequence: Front → Lateral → Behind Relative position 	2.22 m ± 1.16 m	15.06 ± 12.06 km/h 4.18 ± 3.35 m/s	PET-based risk  TTAC-based risk 	Spatial deviation responsiveness  Spatial deviation combinations (ped/cyc)  Speed variation combinations (ped/cyc) 

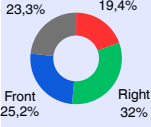
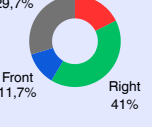
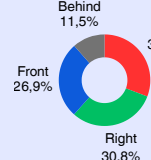
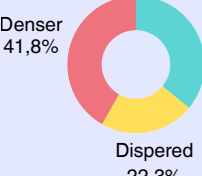
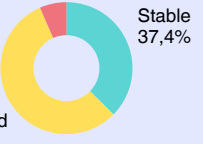
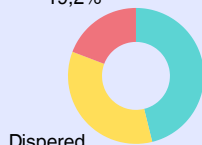
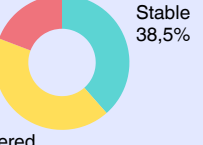
Abbreviations:

- Ped: Pedestrian
- Cyc: Cyclist
- Acc: Acceleration
- Dec: Deceleration
- Cst: Constant

Icon	Class	Direction	Approach angle	Position	Average minimal distance	Average relative speed	Risk metrics	Reactivity statistics
	Give way (38 detected in CTV)	Crossing (intersection of paths)	Approach & Moving away	Relative position Behind 10,5% Left 23,7% Front 42,1% Right 23,7%	3.3 m ± 0.86 m	8.12 ± 2.94 km/h 2.26 ± 0.82 m/s	PET-based risk Low 26,3% Medium 73,7% TTAC-based risk Low Critical 2,8% Medium 22,2% High 61,1% 13,9%	Spatial deviation responsiveness Ped 71% Cyc 29% Spatial deviation combinations (ped/cyc) Cst/Cst 100% Speed variation combinations (ped/cyc) Other 15,4% Acc/Acc 13,6% Acc/Dec 10,5% Dec/Dec 26,3% Dec/Acc 34,2%
	Moving away (539 detected in CTV)	All	Moving away	Relative position Behind 36,7% Left 22,1% Front 19,9% Right 21,3%	3.22 m ± 0.96 m	12.74 ± 4.94 km/h 3.54 ± 1.37 m/s	PET-based risk Low 25% Medium 37,3% High 23% Critical 14,7% TTAC-based risk Low 10,9% Medium 22% High 39,7% Critical 27,5%	Spatial deviation responsiveness Ped 60% Cyc 40% Spatial deviation combinations (ped/cyc) Cst/Dev 0,4% Cst/Cst 99,6% Speed variation combinations (ped/cyc) Other 7,6% Acc/Acc 20,6% Acc/Dec 17,6% Dec/Dec 25,8% Dec/Acc 28,4%

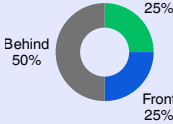
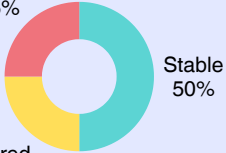
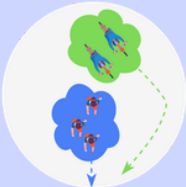
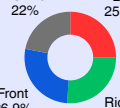
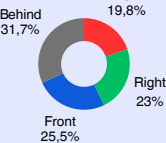
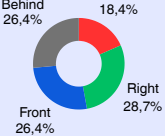
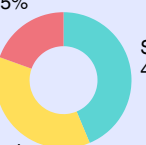
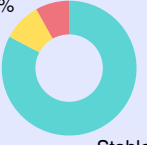
Icon	Class	Direction	Approach angle	Position	Average minimal distance	Average relative speed	Risk metrics	Reactivity statistics
	Following (not observed in CTV videos but 175 detected)	Same	Approach / Lateral / Moving away	Relative position 	3 m ± 1.02 m	12.74 ± 4.94 km/h 3.54 ± 1.37 m/s	PET-based risk  TTAC-based risk 	Spatial deviation responsiveness  Spatial deviation combinations (ped/cyc)  Speed variation combinations (ped/cyc) 

Group-Individual & Group-Group Interactions

Icon	Class	Direction	Approach angle	Position	Average minimal distance	Average relative speed	Group Density Evolution
	Weaving (1 detected in CTV)	All	Approach & Moving away	Sequence: Behind / Front → Lateral → Front / Behind (according to direction) Relative position (GP-1C)  Right 100%	Global: 2.11 m GP-1C: 2.11 m	Global: 5.84 km/h (1.62 m/s) GP-1C: 5.84 km/h (1.62 m/s)	Pedestrians density variation (GP-1C)  Stable 100%
	Group split (145 detected in CTV)	All	Approach & Moving away	Sequence: Behind / Front → Lateral → Front / Behind (according to direction) Relative position (GP-1C)  Relative position (1P-GC)  Relative position (GP-GC) 	Global: 1.75 ± 0.99 m GP-1C: 1.72 ± 1 m 1P-GC: 1.93 ± 1 m GP-GC: 1.75 ± 0.97 m	Global: 6.54 ± 3.06 km/h (1.8 ± 0.85 m/s) GP-1C: 7.18 ± 2.85 km/h (2.08 ± 0.79 m/s) 1P-GC: 5.1 ± 2.53 km/h (1.42 ± 0.70 m/s) GP-GC: 4.89 ± 3.34 km/h (1.36 ± 0.93 m/s)	Pedestrians density variation (GP-1C)  Mean density = 1.86 ± 2.8 Cyclists density variation (1P-GC)  Mean density = 3.31 ± 8.34 Pedestrians density variation (GP-GC)  Mean density = 3.31 ± 5.54 Cyclists density variation (GP-GC)  Mean density = 10.16 ± 20.78

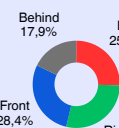
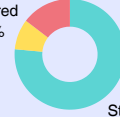
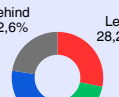
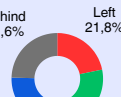
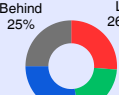
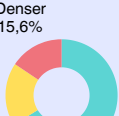
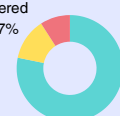
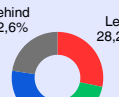
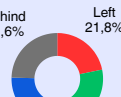
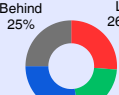
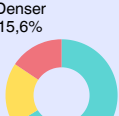
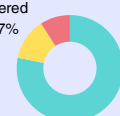
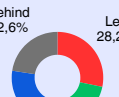
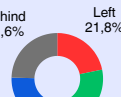
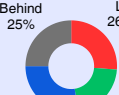
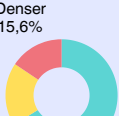
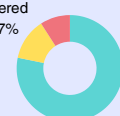
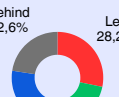
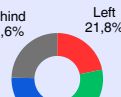
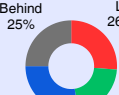
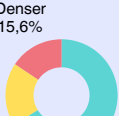
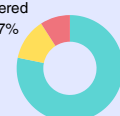
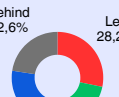
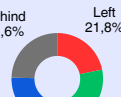
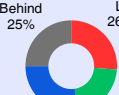
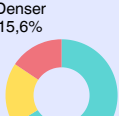
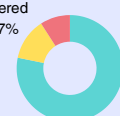
Abbreviations:

- GP-1C: interaction with a pedestrians group and 1 cyclist
- 1P-GC: interaction with 1 pedestrian and a cyclistd group
- GP-GC: interaction with a pedestrians group and a cyclists group

Icon	Class	Direction	Approach angle	Position	Average minimal distance	Average relative speed	Group Density Evolution
	Weaving & Group split (4 detected in CTV)	All	Approach & Moving away	Sequence: Behind / Front → Lateral → Front / Behind (according to direction) Relative position (GP-1C) 	Global: 1.3 ± 0.97 m GP-1C: 1.3 ± 0.97 m	Global: 6.34 ± 1.31 km/h $(1.76 \pm 0.36$ m/s) GP-1C: 6.34 ± 1.31 km/h $(1.76 \pm 0.36$ m/s)	Pedestrians density variation (GP-1C)  Mean density = 0.99 ± 1.15
	Overtaking (739 detected in CTV)	Same	Approach & Moving away	Sequence: Behind → Lateral → Front Relative position (GP-1C)  Relative position (1P-GC)  Relative position (GP-GC) 	Global: 2.43 ± 1.17 m GP-1C: 2.34 ± 1.18 m 1P-GC: 2.7 ± 1.15 m GP-GC: 2.49 ± 1.06 m	Global: 6.01 ± 4.41 km/h $(1.67 \pm 1.23$ m/s) GP-1C: 6.34 ± 4.9 km/h $(1.76 \pm 1.36$ m/s) 1P-GC: 5.72 ± 3.2 km/h $(1.59 \pm 0.89$ m/s) GP-GC: 4.73 ± 2.94 km/h $(1.31 \pm 0.82$ m/s)	Pedestrians density variation (GP-1C)  Mean density = 18.16 ± 99.83 (due to noise) Cyclists density variation (1P-GC)  Mean density = 2.35 ± 8.77 Pedestrians density variation (GP-GC)  Mean density = 10.34 ± 25.46 Cyclists density variation (GP-GC)  Mean density = 1.4 ± 5.73

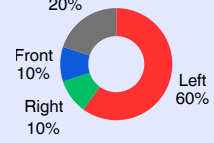
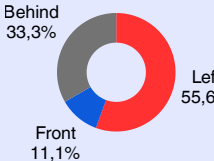
Abbreviations:

- GP-1C: interaction with a pedestrians group and 1 cyclist
- 1P-GC: interaction with 1 pedestrian and a cyclistd group
- GP-GC: interaction with a pedestrians group and a cyclists group

Icon	Class	Direction	Approach angle	Position	Average minimal distance	Average relative speed	Group Density Evolution	
	Bypassing (680 detected in CTV)	Opposite	Approach & Moving away	Sequence: Front → Lateral → Behind Relative position (GP-1C) Behind 23,1% Left 21,8% Front 27,1% Right 28%  Relative position (1P-GC) Behind 22,8% Left 22% Front 26% Right 29,3%  Relative position (GP-GC) Behind 17,9% Left 25,4% Front 28,4% Right 28,4% 	Global: 2.29 ± 1.14 m GP-1C: 2.22 ± 1.11 m 1P-GC: 2.55 ± 1.19 m GP-GC: 2.31 ± 1.19 m	Global: 6.75 ± 3.56 km/h (1.88 ± 0.99 m/s) GP-1C: 6.8 ± 3.77 km/h (1.89 ± 1.05 m/s) 1P-GC: 6.82 ± 2.99 km/h (1.89 ± 0.83 m/s) GP-GC: 6.3 ± 2.39 km/h (1.75 ± 0.66 m/s)	Pedestrians density variation (GP-1C) Denser 14,2% Dispersed 19,3% Stable 66,4% Mean density = 11.08 ± 44.5 (due to noise) 	Cyclists density variation (1P-GC) Denser 14,6% Dispersed 8,9% Stable 76,5% Mean density = 7.17 ± 25.15 
				Sequence: Behind / Front → Lateral → Front / Behind Relative position (GP-1C) Behind 22,6% Left 28,2% Front 20,1% Right 29,1%  Relative position (1P-GC) Behind 24,6% Left 21,8% Front 29,1% Right 24,6%  Relative position (GP-GC) Behind 25% Left 26,2% Front 27,4% Right 21,4% 	Global: 3.03 ± 1.14 m GP-1C: 3.02 ± 1.14 m 1P-GC: 3.14 ± 1.12 m GP-GC: 2.92 ± 1.13 m	Global: 7.23 ± 6.68 km/h (2.01 ± 1.86 m/s) GP-1C: 7.89 ± 6.12 km/h (2.19 ± 1.7 m/s) 1P-GC: 6.33 ± 9.44 km/h (1.76 ± 2.62 m/s) GP-GC: 5.67 ± 3.49 km/h (1.56 ± 0.97 m/s)	Pedestrians density variation (GP-1C) Denser 15,6% Dispersed 18,2% Stable 66,2% Mean density = 12.42 ± 84.1 (due to noise) 	Cyclists density variation (1P-GC) Dispersed 12,7% Stable 78,2% Mean density = 3.85 ± 11.94 
	Crossing (546 detected in CTV)	Crossing (intersection of paths)	Approach & Moving away (obliquely or not)	Sequence: Behind / Front → Lateral → Front / Behind Relative position (GP-1C) Behind 22,6% Left 28,2% Front 20,1% Right 29,1%  Relative position (1P-GC) Behind 24,6% Left 21,8% Front 29,1% Right 24,6%  Relative position (GP-GC) Behind 25% Left 26,2% Front 27,4% Right 21,4% 	Global: 3.03 ± 1.14 m GP-1C: 3.02 ± 1.14 m 1P-GC: 3.14 ± 1.12 m GP-GC: 2.92 ± 1.13 m	Global: 7.23 ± 6.68 km/h (2.01 ± 1.86 m/s) GP-1C: 7.89 ± 6.12 km/h (2.19 ± 1.7 m/s) 1P-GC: 6.33 ± 9.44 km/h (1.76 ± 2.62 m/s) GP-GC: 5.67 ± 3.49 km/h (1.56 ± 0.97 m/s)	Pedestrians density variation (GP-1C) Denser 15,6% Dispersed 18,2% Stable 66,2% Mean density = 12.42 ± 84.1 (due to noise) 	Cyclists density variation (1P-GC) Dispersed 12,7% Stable 78,2% Mean density = 3.85 ± 11.94 
				Sequence: Behind / Front → Lateral → Front / Behind Relative position (GP-1C) Behind 22,6% Left 28,2% Front 20,1% Right 29,1%  Relative position (1P-GC) Behind 24,6% Left 21,8% Front 29,1% Right 24,6%  Relative position (GP-GC) Behind 25% Left 26,2% Front 27,4% Right 21,4% 	Global: 3.03 ± 1.14 m GP-1C: 3.02 ± 1.14 m 1P-GC: 3.14 ± 1.12 m GP-GC: 2.92 ± 1.13 m	Global: 7.23 ± 6.68 km/h (2.01 ± 1.86 m/s) GP-1C: 7.89 ± 6.12 km/h (2.19 ± 1.7 m/s) 1P-GC: 6.33 ± 9.44 km/h (1.76 ± 2.62 m/s) GP-GC: 5.67 ± 3.49 km/h (1.56 ± 0.97 m/s)	Pedestrians density variation (GP-1C) Denser 15,6% Dispersed 18,2% Stable 66,2% Mean density = 12.42 ± 84.1 (due to noise) 	Cyclists density variation (1P-GC) Dispersed 12,7% Stable 78,2% Mean density = 3.85 ± 11.94 
	Crossing (546 detected in CTV)	Crossing (intersection of paths)	Approach & Moving away (obliquely or not)	Sequence: Behind / Front → Lateral → Front / Behind Relative position (GP-1C) Behind 22,6% Left 28,2% Front 20,1% Right 29,1%  Relative position (1P-GC) Behind 24,6% Left 21,8% Front 29,1% Right 24,6%  Relative position (GP-GC) Behind 25% Left 26,2% Front 27,4% Right 21,4% 	Global: 3.03 ± 1.14 m GP-1C: 3.02 ± 1.14 m 1P-GC: 3.14 ± 1.12 m GP-GC: 2.92 ± 1.13 m	Global: 7.23 ± 6.68 km/h (2.01 ± 1.86 m/s) GP-1C: 7.89 ± 6.12 km/h (2.19 ± 1.7 m/s) 1P-GC: 6.33 ± 9.44 km/h (1.76 ± 2.62 m/s) GP-GC: 5.67 ± 3.49 km/h (1.56 ± 0.97 m/s)	Pedestrians density variation (GP-1C) Denser 15,6% Dispersed 18,2% Stable 66,2% Mean density = 12.42 ± 84.1 (due to noise) 	Cyclists density variation (1P-GC) Dispersed 12,7% Stable 78,2% Mean density = 3.85 ± 11.94 
				Sequence: Behind / Front → Lateral → Front / Behind Relative position (GP-1C) Behind 22,6% Left 28,2% Front 20,1% Right 29,1%  Relative position (1P-GC) Behind 24,6% Left 21,8% Front 29,1% Right 24,6%  Relative position (GP-GC) Behind 25% Left 26,2% Front 27,4% Right 21,4% 	Global: 3.03 ± 1.14 m GP-1C: 3.02 ± 1.14 m 1P-GC: 3.14 ± 1.12 m GP-GC: 2.92 ± 1.13 m	Global: 7.23 ± 6.68 km/h (2.01 ± 1.86 m/s) GP-1C: 7.89 ± 6.12 km/h (2.19 ± 1.7 m/s) 1P-GC: 6.33 ± 9.44 km/h (1.76 ± 2.62 m/s) GP-GC: 5.67 ± 3.49 km/h (1.56 ± 0.97 m/s)	Pedestrians density variation (GP-1C) Denser 15,6% Dispersed 18,2% Stable 66,2% Mean density = 12.42 ± 84.1 (due to noise) 	Cyclists density variation (1P-GC) Dispersed 12,7% Stable 78,2% Mean density = 3.85 ± 11.94 

Abbreviations:

- GP-1C: interaction with a pedestrians group and 1 cyclist
- 1P-GC: interaction with 1 pedestrian and a cyclistd group
- GP-GC: interaction with a pedestrians group and a cyclists group

Icon	Class	Direction	Approach angle	Position	Average minimal distance	Average relative speed	Group Density Evolution
	Weak interaction (19 detected in CTV)	All	Moving away	Relative position (GP-1C)  Relative position (1P-GC) 	Global: 3.38 ± 1.17 m GP-1C: 3.46 ± 1.25 m 1P-GC: 3.3 ± 1.13 m	Global: 5.6 ± 4.39 km/h $(1.56 \pm 1.21$ m/s) GP-1C: 3.99 ± 3 km/h $(1.89 \pm 1.05$ m/s) 1P-GC: 7.4 ± 5.14 km/h $(1.89 \pm 0.83$ m/s)	Pedestrians density variation (GP-1C)  Mean density = 1.71 ± 5.39 Cyclists density variation (1P-GC)  Mean density = 2.87 ± 6.57
	Give way (not observed in CTV)	Crossing (intersection of paths)	Approach & Moving away	Lateral	Unknown	Unknown	Unknown

Abbreviations:

- GP-1C: interaction with a pedestrians group and 1 cyclist
- 1P-GC: interaction with 1 pedestrian and a cyclistd group
- GP-GC: interaction with a pedestrians group and a cyclists group

Spatio-temporal metrics for the classification

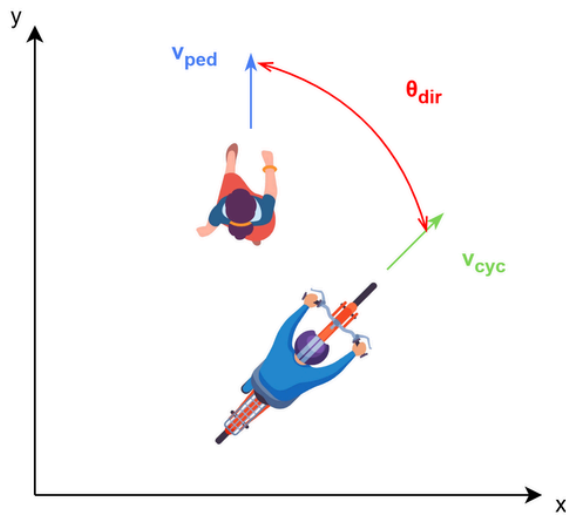
Metric	Definition	Utility	Interpretation	Formula
DBSCAN (Density-Based Spatial Clustering of Applications with Noise)	Cluster detection technique based on: - inter-agent distance (1.5m for pedestrians, 2.5m for cyclists) - direction	- Cyclist and pedestrian group detection. - Detection of individuals outside clusters (noise). - Detection of group splitting (e.g. caused by a passing cyclist).	Cluster: group with at least two nearby agents going in the same direction. Noise: single agent not belonging to any cluster. SPLIT = TRUE: group split detected.	sklearn.cluster : DBSCAN
Convex hull	Smallest convex set containing all agents of a cluster.	Detection of another user type within the hull	IN_HULL = TRUE: weaving of an agent through a group.	scipy.spatial : ConvexHull
Minimum inter-agent distance	Distance between the closest agents of two different types (e.g. two different groups).	- Proximity detection - Risk assessment	< 1.5 m: CRITICAL_PROXIMITY 1.5-3 m: CLOSE 3-5 m: MODERATE_DISTANCE > 5: FAR	$d(t) = \sqrt{(x_p(t) - x_c(t))^2 + (y_p(t) - y_c(t))^2}$ $d_{min} = \min_t d(t)$ $x_p(t), y_p(t)$: coordinates of the pedestrian at time t, $x_c(t), y_c(t)$: coordinates of the cyclist at time t, $d(t)$: inter-agent distance at time t, d_{min} : minimum distance observed during the interaction.
Relative speed	Relative (average) speed of the cyclist(s) with respect to the pedestrian or pedestrian group.	- Analysis of relative-speed evolution during the interaction - Risk assessment	0-3 km/h: VERY_SLOW 3-15 km/h: MODERATE 15-20 km/h: FAST > 20 km/h: VERY_FAST	$\mathbf{v}_{rel} = \mathbf{v}_c - \mathbf{v}_p$ $\mathbf{v}_{rel} = \begin{bmatrix} v_{x,c} - v_{x,p} \\ v_{y,c} - v_{y,p} \end{bmatrix}$ $\mathbf{v}_p$: pedestrian velocity vector, $\mathbf{v}_c$: cyclist velocity vector, $v_{x,p}, v_{y,p}$: x- and y-components of the pedestrian velocity, $v_{x,c}, v_{y,c}$: x- and y-components of the cyclist velocity, $\mathbf{v}_{rel}$: relative velocity vector

Metric	Definition	Utility	Interpretation	Formula
Relative position	- Relative position of the cyclist with respect to the pedestrian. - For groups, relative position of the closest agents. - Relative angle between the pedestrian's direction and the cyclist's direction (as seen from the pedestrian).	Identify the cyclist's position from the pedestrian's point of view during the interaction.	[-45° ; 45°] : FRONT (cyclist in front of pedestrian) [45° ; 135°] : LEFT (cyclist on the left) [135° ; -135°] : BEHIND (cyclist behind pedestrian) [-45° ; -135°] : RIGHT (cyclist on the right)	$\mathbf{r} = \begin{bmatrix} x_c - x_p \\ y_c - y_p \end{bmatrix}$ $\theta_{pos} = \text{atan2}(y_c - y_p, x_c - x_p)$ $\mathbf{r}$: relative position vector from the pedestrian to the cyclist, θ_{pos} : relative bearing angle of the cyclist with respect to the pedestrian, x_p, y_p : coordinates of the pedestrian, x_c, y_c : coordinates of the cyclist, atan2 : four-quadrant inverse tangent function.
Direction angle	- Users' directions. - Angle between the (average) velocity vectors of the pedestrian(s) and the cyclist(s).	Identify agents' movement directions during the interaction.	[0° ; 30°]: SAME_DIRECTION [30° ; 75°]: DIVERGING [75° ; 120°]: CROSSING [120° ; 150°]: DIVERGING [150° ; 180°]: OPPOSITE_DIRECTION	$\theta_{dir} = \arccos \left(\frac{\mathbf{v}_p \cdot \mathbf{v}_c}{\ \mathbf{v}_p\ \ \mathbf{v}_c\ } \right)$ θ_{dir} : direction angle, $\mathbf{v}_p$: pedestrian velocity vector, $\mathbf{v}_c$: cyclist velocity vector, $\mathbf{v}_p \cdot \mathbf{v}_c$: dot product of the velocity vectors, $\ \mathbf{v}_p\$: norm of the pedestrian velocity vector, $\ \mathbf{v}_c\$: norm of the cyclist velocity vector.
Approach angle	Angle between the pedestrian's movement direction and the direction from which the cyclist approaches.	Identify how the cyclist approaches the pedestrian.	[0° ; 30°]: FRONTAL_APPROACH (cyclist heading straight toward the pedestrian) [30° ; 75°]: OBLIQUE_APPROACH [75° ; 105°]: LATERAL_APPROACH (or crossing) > 105°: MOVING_AWAY (cyclist moving away from the pedestrian)	$\theta_{app} = \arccos \left(\frac{\mathbf{v}_{rel} \cdot \mathbf{r}}{\ \mathbf{v}_{rel}\ \ \mathbf{r}\ } \right)$ θ_{app} : approach angle, $\mathbf{r}$: relative position vector, $\mathbf{v}_{rel}$: relative velocity vector, $\mathbf{v}_{rel} \cdot \mathbf{r}$: dot product between the relative velocity and relative position vectors, $\ \mathbf{v}_{rel}\$: norm of the relative velocity vector, $\ \mathbf{r}\$: norm of the relative position vector.

Metric	Definition	Utility	Interpretation	Formula
PET (Post-Encroachment Time)	Time difference between the moment one user leaves the conflict zone (trajectory intersection) and the moment the second user enters it.	- Measures the temporal gap that would separate two users from colliding. - Post-interaction risk evaluation (surrogate safety measure).	< 1 s: CRITICAL (near-collision) 1-2 s: HIGH (high risk of collision) 2-4 s: MEDIUM > 4 s: LOW	$PET = t_{entry,2} - t_{exit,1}$ $t_{exit,1}$: time at which the first road user leaves the conflict area, $t_{entry,2}$: time at which the second road user enters the conflict area.
TTAC (Time-To-Avoided-Collision-Point)	Remaining time for the agents to avoid each other. Computed throughout the interaction, assuming constant directions and speeds.	Minimal TTAC: interaction-severity indicator.	< 1 s: CRITICAL (near-collision) 1-2 s: HIGH (high risk of collision) 2-4 s: MEDIUM > 4 s: LOW	$TTAC(t) = \max(t_{ped}, t_{cyc})$ t_{ped} : predicted time for the pedestrian to reach the conflict point, t_{cyc} : predicted time for the cyclist to reach the conflict point, $\max(t_{ped}, t_{cyc})$: arrival time of the second agent reaching the conflict point.

Direction angle (θ_{dir})

Angle between the two velocity vectors



$$\theta_{dir} = \arccos \left(\frac{\mathbf{v}_p \cdot \mathbf{v}_c}{\|\mathbf{v}_p\| \|\mathbf{v}_c\|} \right)$$

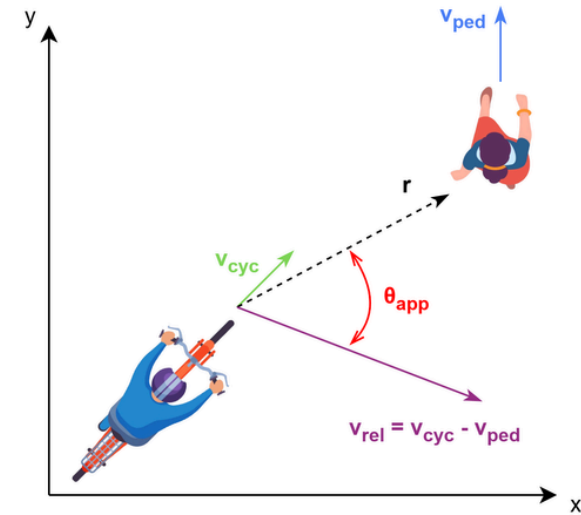
-  $\mathbf{v}_{cyc}$: cyclist velocity vector
-  $\mathbf{v}_{ped}$: pedestrian velocity vector
-  θ_{dir} : direction angle

Interpretation

Same direction	Opposite direction	Crossing (perpendicular)	Diverging / Converging
$\theta_{app} = 0^\circ$	$\theta_{app} = 180^\circ$	$\theta_{app} = 90^\circ$	$0^\circ < \theta_{app} < 180^\circ$
			

Approach angle (θ_{app})

Angle between the relative velocity vector and the relative position vector from cyclist to pedestrian



$$\theta_{app} = \arccos \left(\frac{\mathbf{v}_{rel} \cdot \mathbf{r}}{\|\mathbf{v}_{rel}\| \|\mathbf{r}\|} \right)$$

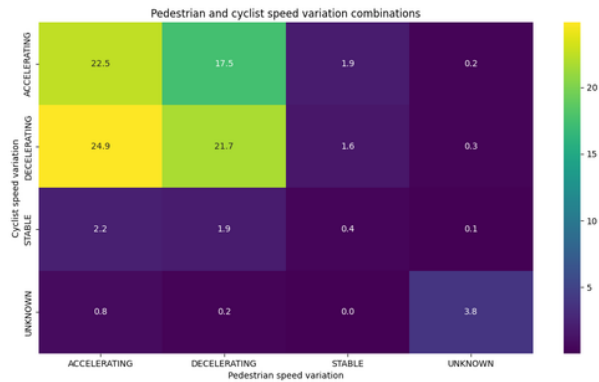
-  $\mathbf{v}_{cyc}$: cyclist velocity vector
-  $\mathbf{v}_{ped}$: pedestrian velocity vector
-  $\mathbf{v}_{rel}$: relative velocity vector
-  $\mathbf{r}$: relative position vector (cyclist w.r.t. pedestrian)
-  θ_{app} : approach angle

Interpretation

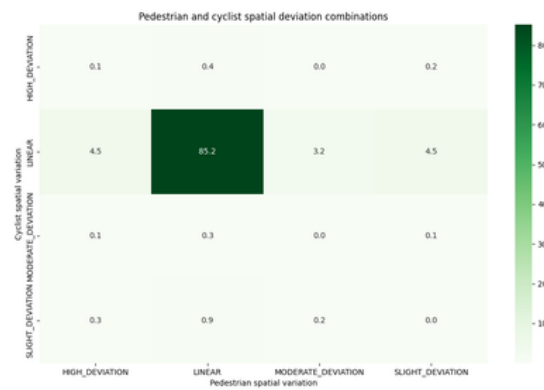
Frontal approach	Crossing / lateral	Moving away
$\theta_{app} = 0^\circ$	$\theta_{app} = 90^\circ$	$\theta_{app} = 180^\circ$
		

Global Results on CTV

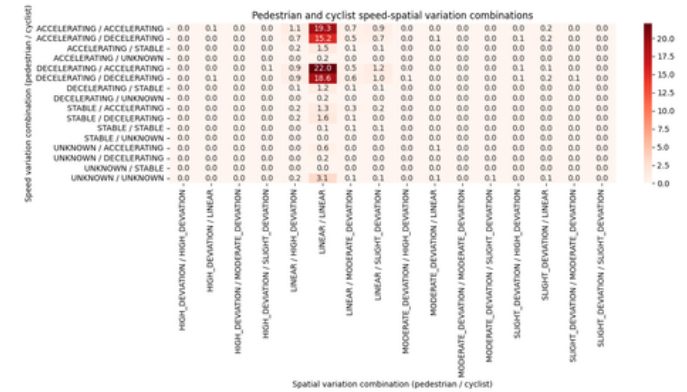
Speed variation combinations



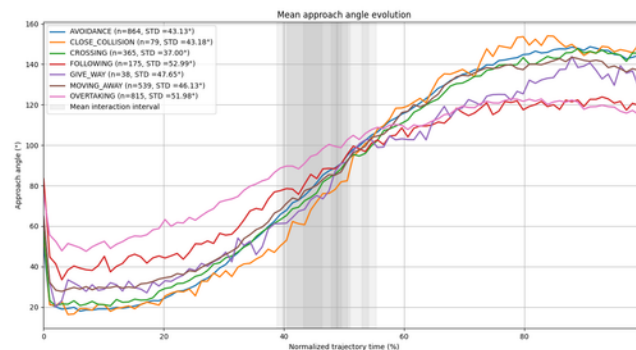
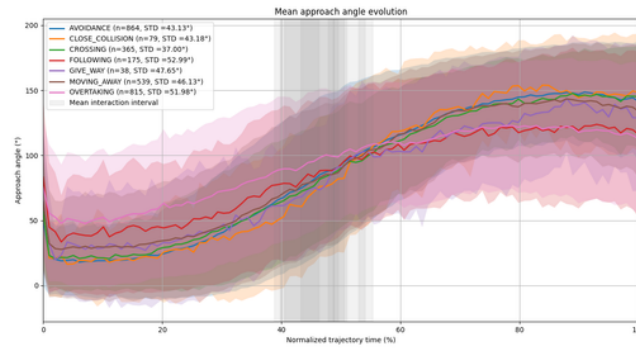
Spatial deviation combinations



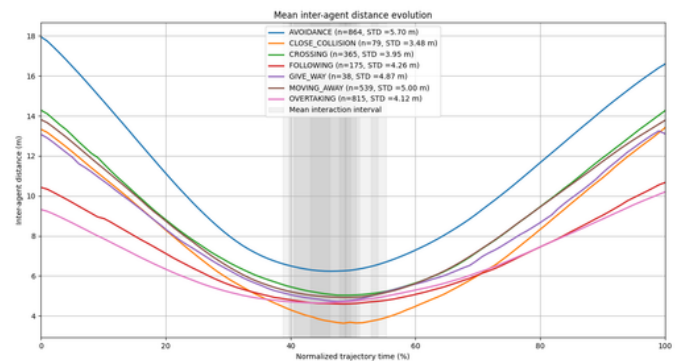
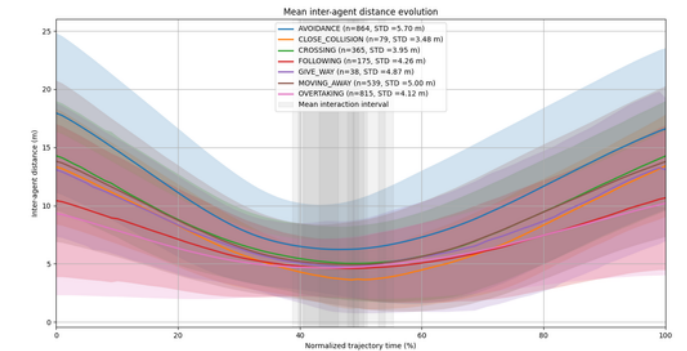
Speed variation and spatial deviation combinations



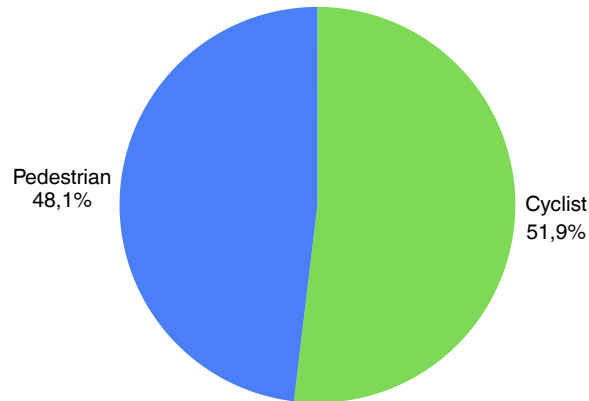
Mean approach angle for all individual-individual interactions



Mean distance for all individual-individual interactions



Global distribution of the most reactive agent spatially for all individual-individual interactions



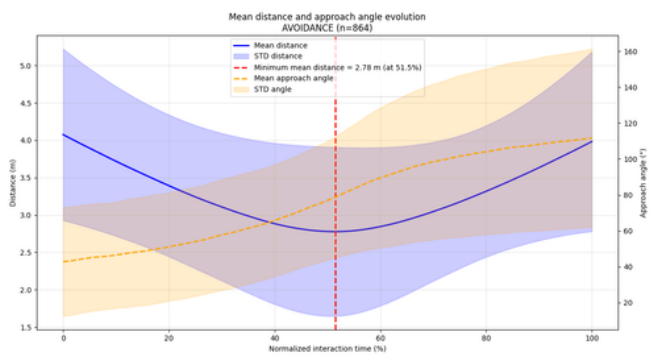
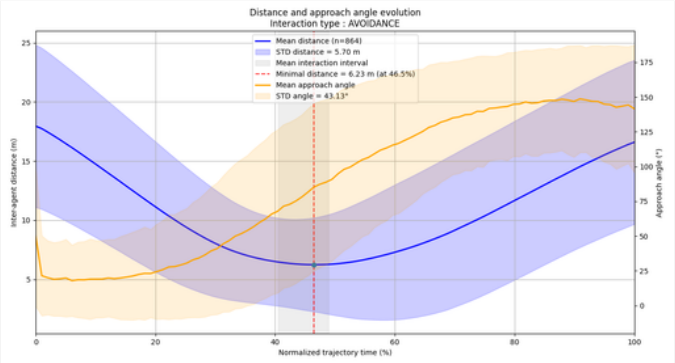
Mean approach angle and mean distance for each individual-individual interactions on CTV

Interaction

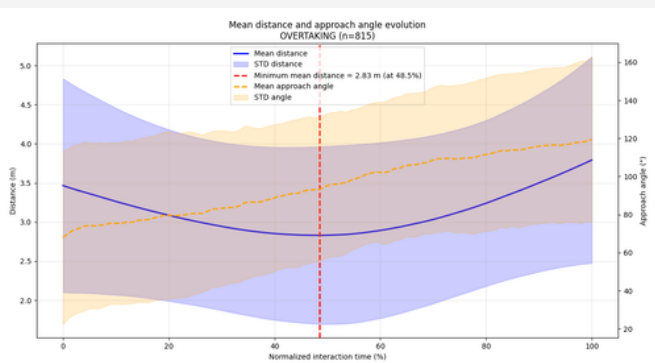
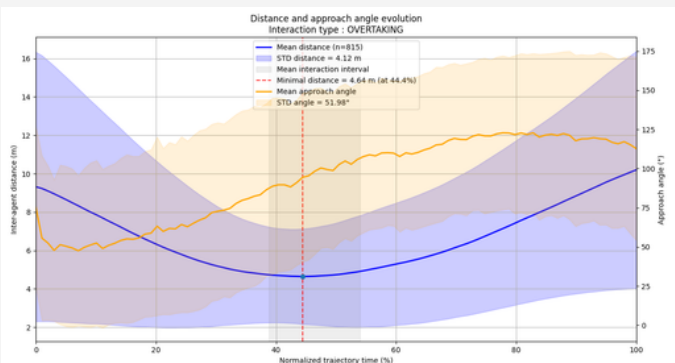
Normalized Trajectory Time

Normalized Interaction Time

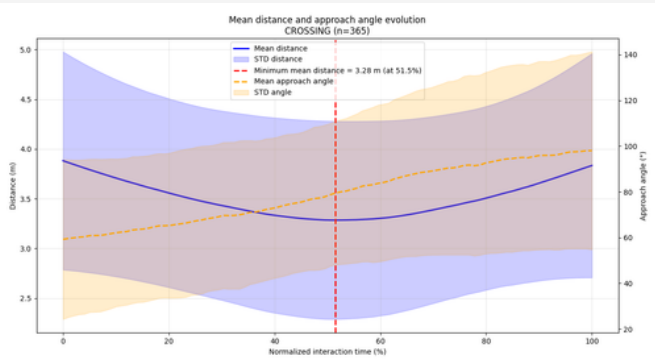
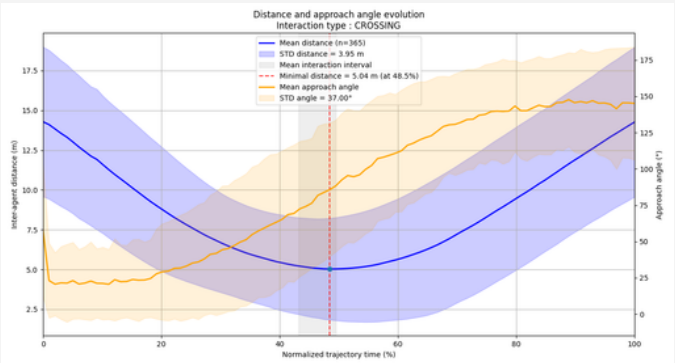
Avoidance



Overtaking



Crossing



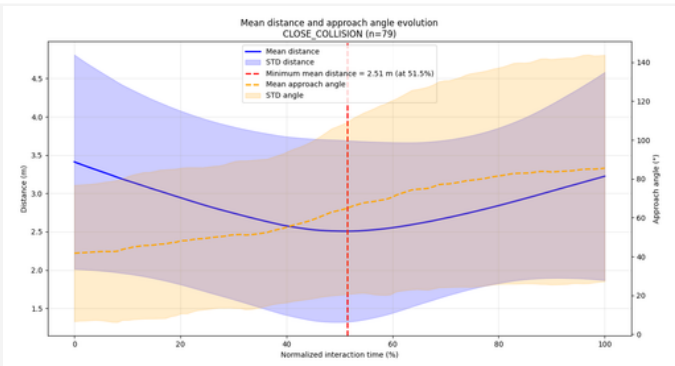
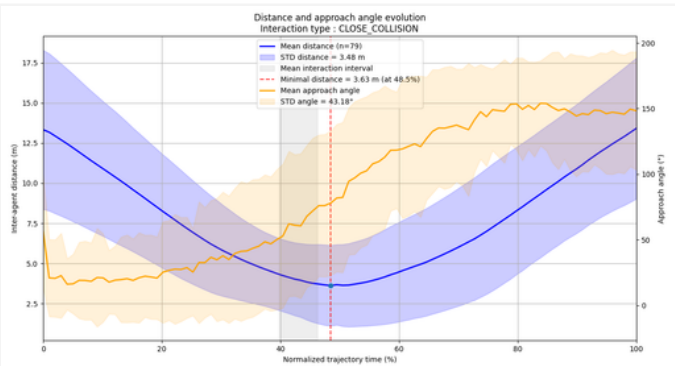
Mean approach angle and mean distance for each individual-individual interactions on CTV

Interaction

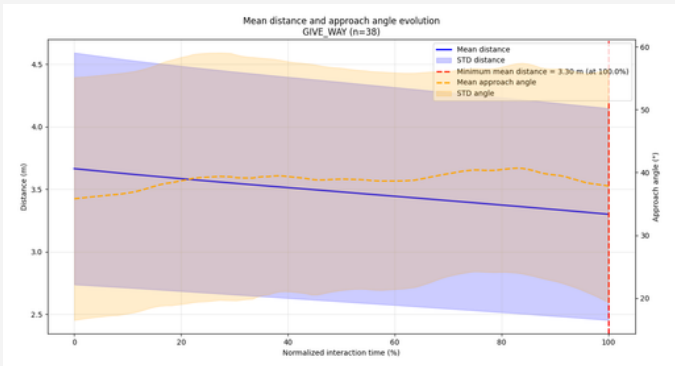
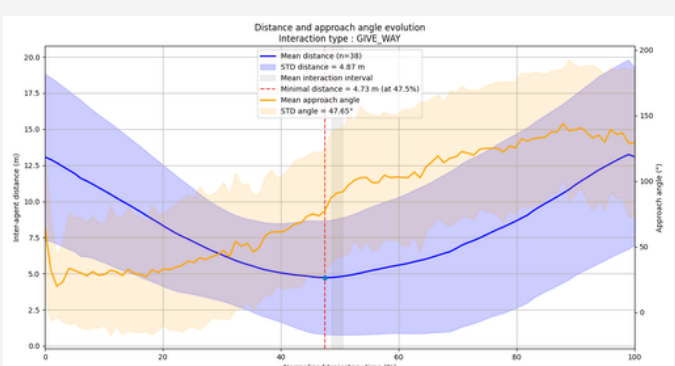
Normalized Trajectory Time

Normalized Interaction Time

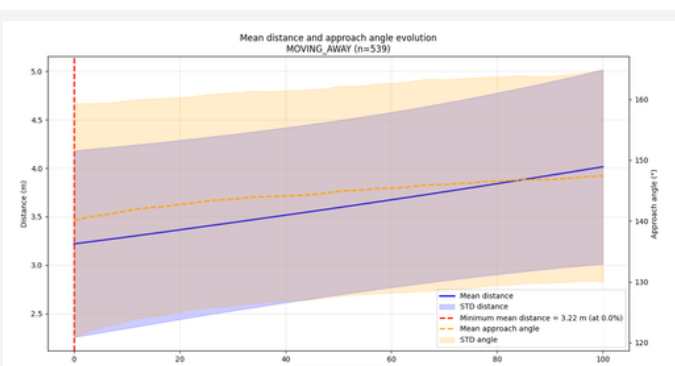
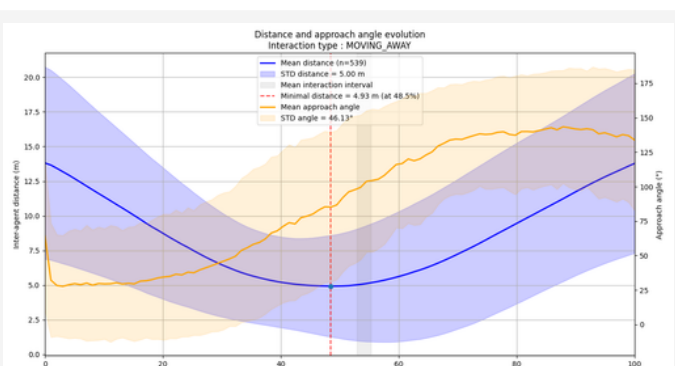
Close-collision



Give way



Moving away



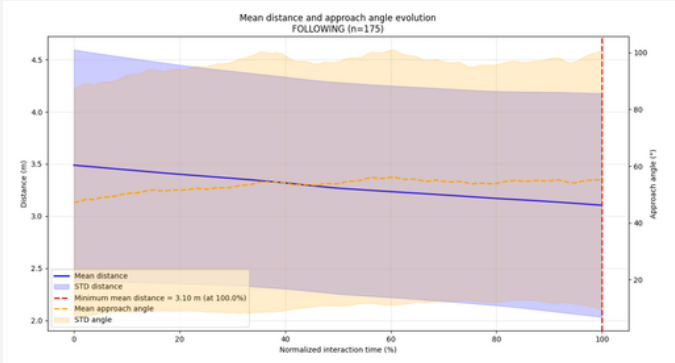
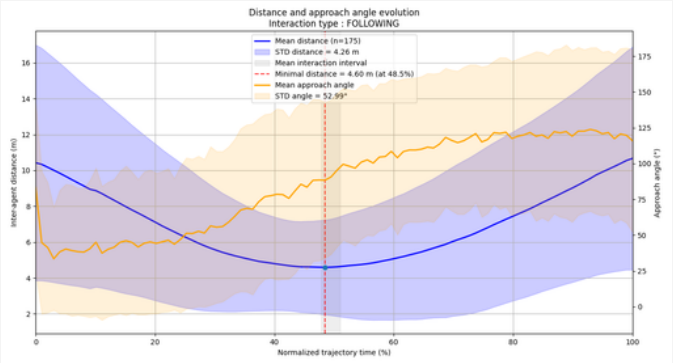
Mean approach angle and mean distance for each individual-individual interactions on CTV

Interaction

Normalized Trajectory Time

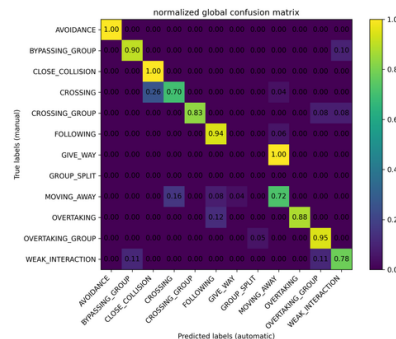
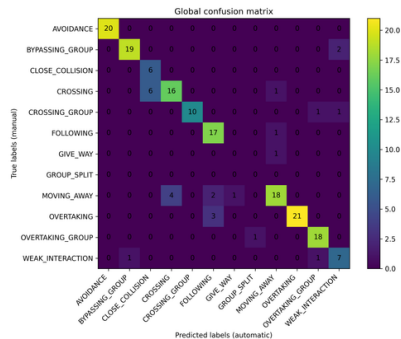
Normalized Interaction Time

Following



Classification validation on TSS, inD and SDD

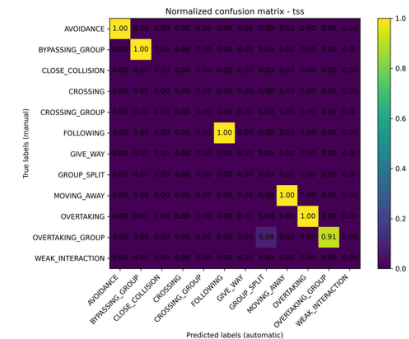
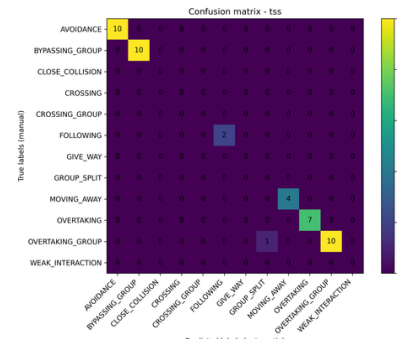
For TSS, inD and SDD (178 manually classified interactions)



Metric	Macro	Weighted
Accuracy	0.85	0.85
Precision	0.71	0.88
Recall	0.72	0.85
F1-score	0.71	0.86

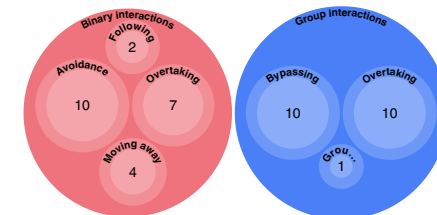
- **Satisfying overall accuracy:** most interactions are correctly classified.
- **Macro scores:** heterogeneous performance across classes.
- **Weighted scores:** strong results for the most frequent classes.

For TSS (44 manually classified interactions)

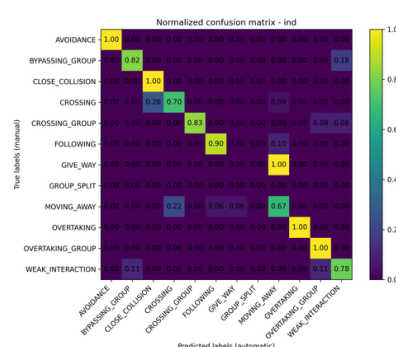
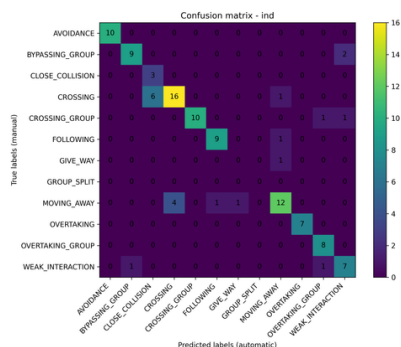


Metric	Macro	Weighted
Accuracy	0.98	0.98
Precision	0.50	1.0
Recall	0.49	0.98
F1-score	0.50	0.99

Selected interactions for manual annotation

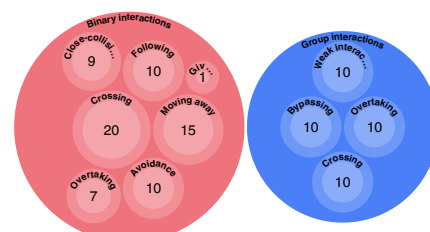


For inD (112 manually classified interactions)

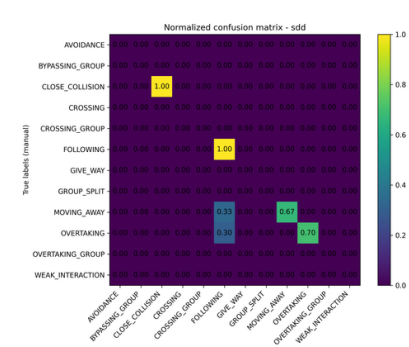
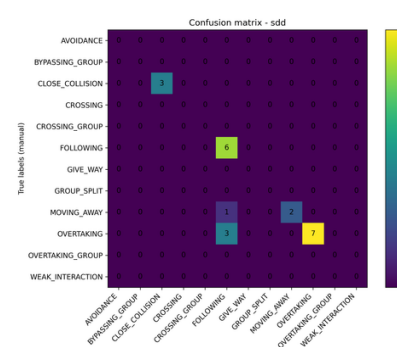


Metric	Macro	Weighted
Accuracy	0.81	0.81
Precision	0.69	0.84
Recall	0.72	0.81
F1-score	0.69	0.82

Selected interactions for manual annotation

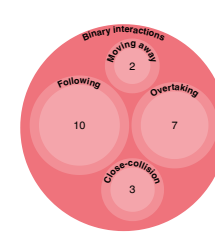


For SDD (22 manually classified interactions)



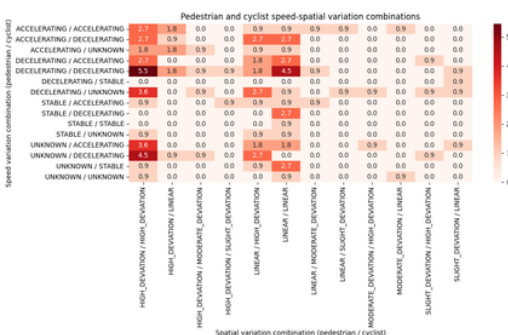
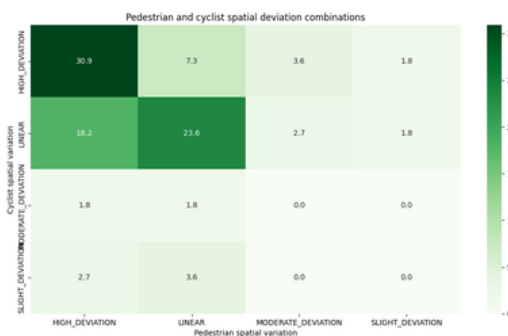
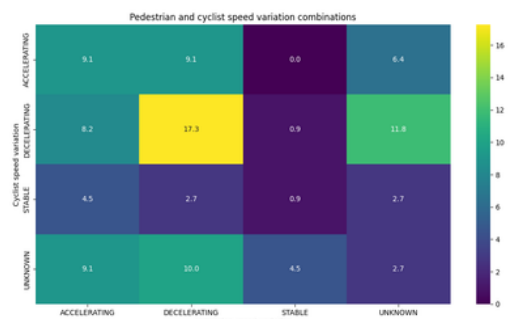
Metric	Macro	Weighted
Accuracy	0.82	0.82
Precision	0.30	0.89
Recall	0.28	0.82
F1-score	0.28	0.82

Selected interactions for manual annotation

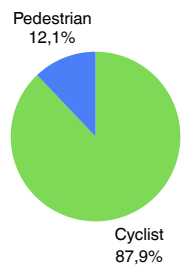


Classification results for TSS, inD and SDD

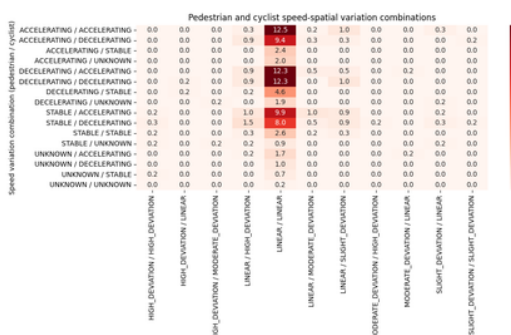
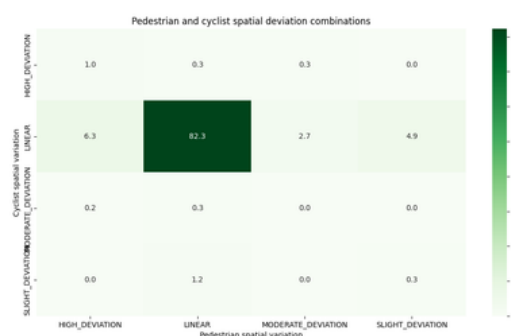
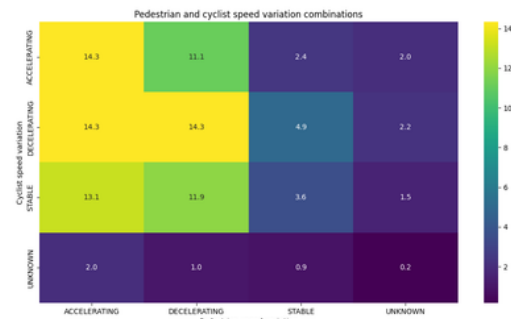
TSS



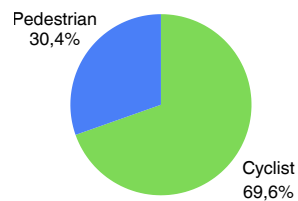
Global distribution of the most reactive agent spatially for all individual-individual interactions



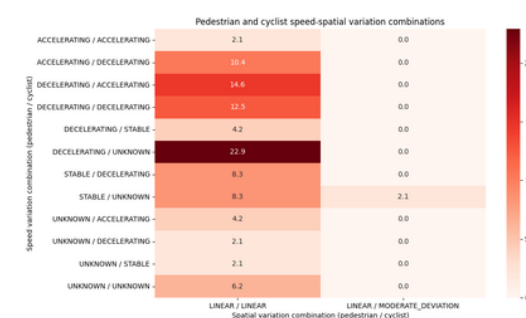
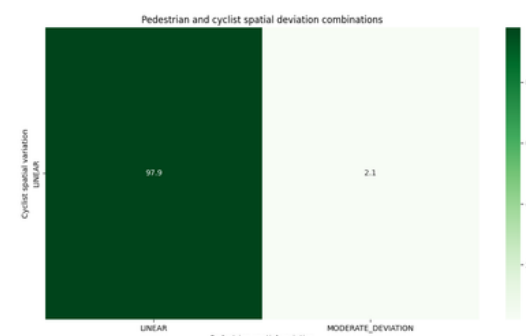
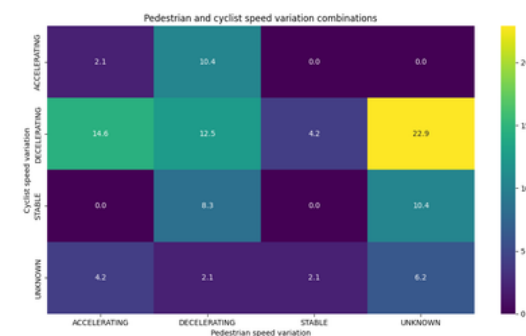
inD



Global distribution of the most reactive agent spatially for all individual-individual interactions



SDD



Global distribution of the most reactive agent spatially for all individual-individual interactions

